

Test 2 Review

Math 252

How to use this review sheet

You don't have to do these in order. Feel free to jump to the types of problems you want to review.

Bonus question: What's missing? Are there any topics covered on the test that don't appear in the questions below?

Want more review problems? See the problems on the homework sheet that you were not required to turn in.

Some Review Problems

1. Committee. Suppose that there are 9 faculty members in the mathematics department and 11 in the computer science department. How many ways are there to select a committee to develop a discrete mathematics course at a school if the committee is to consist of three faculty members from the mathematics department and four from the computer science department?

2. Lottery. In a lottery game, players select 6 distinct numbers from the numbers 1 – 40. (The order in which they are chosen doesn't matter – often you select the numbers by punching out tabs for the 6 numbers you want and there is no record of the order in which you punched them out.) Later, the lottery commission picks 6 numbers. Compute the probability of each of these events and put the results in a nice table.

- all 6 numbers match.
- exactly 5 of the 6 match.
- exactly 4 of the 6 match.
- exactly 3 of the 6 match.
- exactly 2 of the 6 match.
- exactly 1 of the 6 match.
- none of the 6 match.

What is the expected number of matches?

3. Events A and B are independent, $P(A) = 0.3$, $P(B) = 0.4$.

- Is it possible for A and B to be mutually exclusive? Why or why not?
- What is $P(A \text{ and } B)$?
- What is $P(A \text{ or } B)$?
- What is $P(A | B)$?
- What is $P(B | A)$?

4. Events A and B are mutually exclusive, $P(A) = 0.3$, $P(B) = 0.4$.

- Is it possible for A and B to be independent? Why or why not?
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- What is $P(A | B)$?
- What is $P(B | A)$?

5. $P(A) = 0.5$, $P(B) = 0.7$, and $P(A \text{ and } B) = 0.3$.

- What is $P(A \text{ or } B)$?
- What is $P(A | B)$?
- What is $P(B | A)$?
- Is it possible for A and B to be independent? Why or why not?
- Is it possible for A and B to be mutually exclusive? Why or why not?

6. **Boxes of marbles.** Cheryl has 3 boxes labeled A, B, and C.

- In box A are 4 red marbles and 1 blue marbles.
- In box B are 5 red marbles and 5 blue marbles.
- In box C are 2 red marbles and 8 blue marbles.

Cheryl rolls a die. If she rolls a 1, she randomly selects a marble from box A. If she rolls a 2 or a 3, she randomly selects a marble from box B. If she rolls a 4, 5, or 6, she randomly selects a marble from box C.

- What is the probability that she selects a red marble? [Why is the answer NOT 11/25?]
- If she selects a red marble, what is the probability that it came from box A?
- If she selects a red marble, which box is it most likely to have come from?

7. **Four suits?** A hand of 5 cards is dealt from a standard 52-card deck. What is the probability that it contains at least one card from each of the 4 suits?

8. **Four suits again.** A hand of 13 cards is dealt from a standard 52-card deck. What is the probability that it contains at least one card from each of the 4 suits?

9. **3-of-a-kind.** A hand of 5 cards is dealt from a standard 52-card deck. What is the probability that it is a 3-of-a-kind?

A 3-of-a-kind has 3 cards of the same number and two other cards that are different from the three and from each other. Example: $3\heartsuit 3\spadesuit 3\clubsuit 7\heartsuit J\heartsuit$

10. **Full house.** A hand of 5 cards is dealt from a standard 52-card deck. What is the probability that it is a full house?

A full house has 3 cards with one number and 2 cards with a different number. Example: $3\heartsuit 3\spadesuit 3\clubsuit J\heartsuit J\clubsuit$

11. **Another box of marbles.** A box has 10 red marbles and 5 blue marbles. You randomly select 3 marbles without replacement. Let X = the number of blue marbles selected. What is $E(X)$?

12. **Can you do this 3 different ways?** Redo the previous problem if the marbles are drawn one at a time and replaced in the box before drawing again. Note: You should be able to do this problem three different ways.

13. Grammar 1. Consider the following grammar with nonterminals A , B , and S , start symbol S , and terminals 0 and 1.

$$\begin{aligned} S &\rightarrow \lambda \\ S &\rightarrow AB \\ A &\rightarrow aB \\ A &\rightarrow b \\ B &\rightarrow Ab \\ B &\rightarrow aa \end{aligned}$$

- Is it type 1 (context sensitive)?
- Is it type 2 (context free)?
- Is it type 3 (regular)?
- Generate 3 strings of length at least 4 using this grammar.

14. Grammar 2. Consider the following grammar with nonterminals A , B , and S , start symbol S , and terminals 0 and 1.

$$\begin{aligned} S &\rightarrow \lambda \\ S &\rightarrow AB \\ aAa &\rightarrow aba \\ A &\rightarrow bB \\ A &\rightarrow Ba \\ bBb &\rightarrow bbb \\ B &\rightarrow Ab \end{aligned}$$

- Is it type 1 (context sensitive)?
- Is it type 2 (context free)?
- Is it type 3 (regular)?
- Generate 3 strings of length at least 4 using this grammar.
- Is it possible to generate string with no a's in it?
- Is it possible to generate a string with no b's in it?

15. Give an example of a grammar that is type 2 (context free) but not type 1 (context sensitive). Show that the language generated by this grammar *is* context sensitive by constructing a context sensitive grammar that generates the same language.